String periods

Given a string X of length n, an integer p ($1 \leq p < n$) is called a *period length* of X if $X[i] = X[i+p]$ for all i for which both i and $i+p$ are valid indices into X.

The smallest of X's period lengths is called its *regularity*. If X has no period lengths, its regularity is n.

For example, the period lengths of `abcabcab` are 3 and 6, and the period lengths of `aabaa` are 3 and 4. Hence, the strings `abcabcab` and `aabaa` both have regularity 3. The string `abcb` has no period lengths and therefore its regularity is 4.

Statement

On the ground someone painted n spots in a row, with exactly 1 step between adjacent spots. There are n people, one standing on each spot. Each person has a t-shirt with a letter, so together (when we read their t-shirts left to right) they are forming a string. You are given the string S obtained this way.

The people want to rearrange themselves to produce a new string T with the smallest possible regularity. Additionally, they want to do this as efficiently as possible.

The rearrangement will be performed as a series of actions. Each action must look as follows:

- Each action consists of one person taking either one step left or one step right.
- After each action, each person must be standing at one of the spots.
- After each action, there can be **at most three** people standing on each individual spot – the spots aren't big enough to hold more than three people at the same time.

After the last action the people must again occupy all n spots, with exactly one person per spot. When read left to right, their t-shirts must produce the desired string T.

Find the smallest possible regularity r they can achieve this way, and then the smallest possible number a of actions needed to rearrange the people into some string T with regularity r.

Input format

The first line of the input file contains the number t of test cases. The specified number of test cases follows, one after another.

Each test case consists of two lines. The first line contains the number n of people, the second line the string S initially formed by the letters on their t-shirts.

Each character in each S will be an uppercase or a lowercase English letter (A-Z, a-z).

Output format

For each test case output a single line of the form "r a".

Subproblem W1 (14 points, public)

Input file: W1.in (/static/online2025/W1.b1e3e20ff7e2aa67924995f6ee8edbd5.in)

In this subproblem $t \leq 100$ and in each test $n \leq 10$.

Subproblem W2 (28 points, secret)

Input file: W2.in (/static/online2025/W2.6b40d32e20a3d06f99fd1af10a1fe49d.in)

In this subproblem $t \le 100$ and in each test $n \le 100$ and $r \le 10$. That is, in this subtask it is guaranteed that each S was chosen so that there is a corresponding T with a small regularity.

Subproblem W3 (46 points, secret)

Input file: W3.in (/static/online2025/W3.e78767c6e439579a40a7c3bec95b5d56.in)

In this subproblem $t \le 100$ and in each test $n \le 100$.

Subproblem W4 (12 points, secret)

Input file: W4.in (/static/online2025/W4.e58900595a7df6234824e8a2e5462d49.in)

In this subproblem $t \le 100$ and in each test $n \le 1000$.

Example

input

```
6
8
abcabcab
8
abcabcba
8
abcabcac
5
abcdd
10
aaaaabbbbb
4
PQrS
```

output

```
3 0
3 2
3 4
4 6
2 20
4 0
```

Explanations:

- Test case 1: The input string already has a period length 3 and it is not possible to rearrange it into a string with a shorter period length.
- Test case 2: Last person takes a step left and then the other person standing there takes a step right. This produces the string from the previous test case.
- Test case 3: Two pairs of neighbors can swap places to produce the string `acbacbac`.
- Test case 4: Produce the string `dabcd`. (For fun, note that if we weren't limited to staying in the original n spots, there would be a faster way to produce this string: the rightmost person could take five steps left.)

- Test case 5: Here the optimal solution is for the people to rearrange themselves to form the string `ababababab`.
- Test case 6: Like in test case 1, we cannot improve the string's regularity, so the best solution is to do nothing. Note that both upper- and lowercase letters can appear in the input.